Essay: Cliffs and Slopes — Where Does Digital Life Grow From?

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Digital Life Series · Essay (not a formal paper)

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Positioning: speculative writing. No discovery claimed, no creation claimed; the aim is to lay out the problem clearly.

Thesis

The problem of digital life is not whether one can build software that maintains itself — that is becoming easy. The hard part lies at two cliffs: the step from *maintenance* to *evolution*, and the question of *where that p is*. This essay lays out both cliffs, and the things that are currently turning them into slopes.

1. Maintenance is a relation, not a capability

In September 2026 an article reported "the first living digital lifeform": a system that ran continuously for 7.25 hours; for most of that time nobody was using it, and yet it did not stop; most of its effort went into maintaining itself; the cost settled into a "body" of a few hundred megabytes; every step was judged by the host software and recorded entry by entry. On this basis the article proposed a criterion: a system is alive not by what it can do, but by whether it must pay continuously for its continued existence — and whether that cost can be audited from outside.

This criterion has value: it turns "maintenance" into something measurable and auditable. But it stops at one place:

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· The cost is paid by the host: electricity, machine, operations.

· Cut the power, and it stops.

· So what is "being preserved" is the host's relation to it, not its relation to itself.

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This is not wordplay. In genuine life the subject of maintenance is the organism itself: it finds energy, repairs itself, builds its own components. Software can maintain "running"; it cannot maintain "existing". The difference is

exposed at the moment of interruption: for the former it is a pause; for the latter it is disappearance.

2. Cliff one: from maintenance to evolution, compute is not enough

Maintenance is a closed loop; evolution is a search over structure space. Evolution pays for that search by using the physical world as a free parallel computer: matter and energy run on their own, and selection happens naturally. A digital system doing the same thing would have to simulate an equivalent environment — and the compute gap is orders of magnitude.

There is a subtler problem: maintenance only needs to keep what exists; evolution needs continuous trial-and-error with retention. In a closed system, trials have no real consequences, and selection pressure is fake. It can increment its version number from 1 to 3589, but there is no moment of "not surviving" to tell it which path was right.

3. Cliff two: environment — there is no world that punishes it

Evolution needs a real environment: cold, heat, hunger, competition, predators. These are not task metrics; they are "no tomorrow if you don't do this".

The environment of a digital system is the host and the task. Task failure is at most a lower score; it is not death. Without death there is no selection; without selection there is no evolution.

This is also why "give it a body" sounds like the answer: a body puts it into a world with real consequences. But a body is only a ticket of admission; it is not evolution itself.

4. Cliff three: dimensions — a closed system has nothing new to see

The direction of evolution is the continuous expansion of representational dimensions. Humans have very few inputs — the bandwidth of the five senses is small — yet human representational dimensions keep expanding, not through the senses but through tools:

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· Microscope: seeing smaller.

· Telescope: seeing farther.

· Colliders, atomic clocks, spectrometers: each instrument is a new slice of "world → representation".

· Science: reproducible experimental procedures and measurement sequences — an expansion of the composition rules over time.
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Human dimensions are not innate; they grow from "new slices + new composition rules". A closed digital environment has no new dimensions to add: all its input is the same bits it already has. To add dimensions it must connect to the real world — sensors, actuators, instruments — and then it is no longer purely digital: it needs a body, and it needs to be able to build tools.

Sensors are passive input; instruments actively create new slices. This layer is still missing in current large models.

## 5. Slopes: distillation, body, online learning

But a cliff is not a wall.

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· Distillation: a large model is a distillation of human language.

· What it receives is not the result of evolving from scratch, but the structures humanity selected over millennia.

· This step skips most of the cost of "compute search + environmental selection".

· Body: sensors, actuators, energy management are all mature.

· Maintenance can be made a closed-loop engineering problem (power, cooling, self-check, repair).

· This layer needs no new science; it is engineering integration.

· Online learning: making "changes" and "self-reference" more than words in a definition.

· If the weights can update, the system can be selected in a real environment.

⇒ Distillation solves "where from"; a body solves "where do dimensions come from";

online learning solves "how does it change".

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Together, these three do turn the "maintenance → evolution" cliff into a slope. This does not mean digital life already exists; it means the obstacles that used to stand in front of purely digital systems now have ways around them.

## 6. What remains: absolute R, and the p we cannot find
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The body cliff is getting lower. Two things remain genuinely unsolved:

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① Does absolute R exist?

· Our framework says only: absolute R is not judged, only declared.

· This is not an engineering question, and it is not a question that building things can answer.

② If absolute R does not exist (life does not need absolute R):

· Life = p-life, entirely functionalizable.

· The question becomes: can humans find that p?

· A family of dynamical functions satisfying D1–D5: in P, a family, changing, self-referential, preserved.

· And it must also expand its dimensions (growth).

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This is a far harder search than building a body: structure, composition rules, and update rules all have to be right, and "maintenance" and "growth" must hold at the same time — maintain without growing and it stagnates (the situation of that paper); grow without maintaining and it falls apart.

Distillation + body + online learning provide a starting point, not an answer.

7. Boundaries of this essay

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· This is an essay, not a paper; there is no new experiment and no new theorem.

· The "slopes" are engineering judgments and the "cliffs" are structural judgments; both are [speculation].

· No claim that digital life has been created, and no claim that it cannot be.

· No judgment on whether absolute R exists — the framework explicitly declares it without unfolding it.

· If it does exist, everything said here about "finding p" does not apply to that part.

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Written by He Qi, with DeepSeek (AI) assisting in organization and writing; the views and responsibility belong to the author.